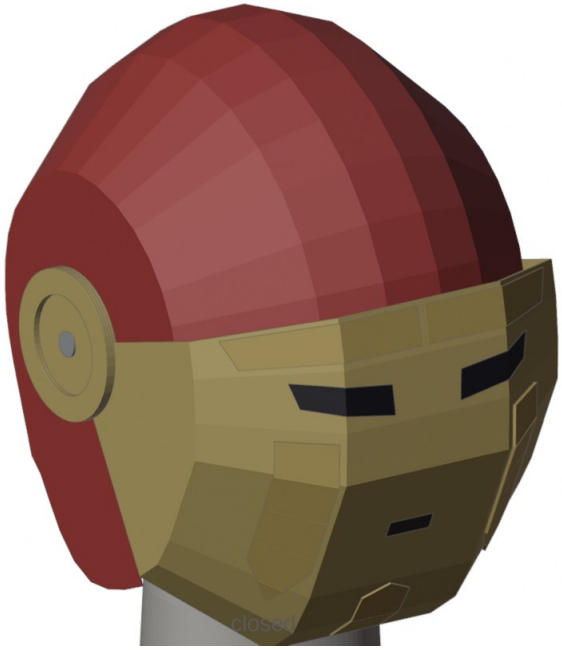
CARDBOARD HELMET

Flip-up faceplate on ear pivots | numbered glue tabs | full-size stencils from a 3D model

Fitted to: forehead 55.0 cm around, chin-over-top-to-nape 66.0 cm, 1.5 mm board. Outside: about 210 mm wide.





1. CHECK THE PRINT FIRST

Print every page on US Letter at 100% / Actual size (turn off 'fit to page'). Measure the bar at the bottom of any page: it must be exactly 100 mm. If it isn't, fix the printer setting before cutting anything.

2. WHAT YOU NEED

- 1.5 mm chipboard (mat board / sketch-pad backing): 11 Letter-size pieces
- Glue stick or spray adhesive (to stick the printouts on)
- PVA or hot glue for the tabs; craft knife, steel ruler, cutting mat
- Empty ballpoint pen for scoring
- 2 Chicago screws (binding posts), flat heads: 5 mm diameter, 6 mm post (shorter than the 7 mm they clamp, so they can be tightened)
- 2 washers: 5 mm hole, 1 mm thick
- Optional: thin foam for padding (up to 8 mm keeps the fit)

3. CHECKED ON THE 3D MODEL BEFORE PRINTING

PASS	Head fit: at least 12.0 mm between your head and the inside of the helmet (target 12); 17 mm above the crown
PASS	Ears: 5.0 mm from the side plates
PASS	Face: faceplate at least 14.8 mm from your face (target 15; nose 15, chin 16 mm)
PASS	Faceplate swing: tested at 161 angles from closed to 80 deg: no contact; tightest gap 1.0 mm; never closer than 13 mm to your head
PASS	Vision: both eyes look straight out through the slots; each eye sees -15 to 15 deg sideways and 0 to 10 deg up/down
PASS	No gaps: 1728 sight lines from inside all hit the helmet
PASS	Putting it on: a collision-free path exists to lower it onto your head (faceplate open)
PASS	Flat panels: every panel flat to 4e-14 mm; normals consistent
PASS	Pieces match the model: every flattened panel within 6e-14 mm of its 3D shape; 236 seams each cut on both sides with matching lengths; 234 glue tabs
PASS	Pages: 11 stencil pages; every piece inside the printable area, none touching
PASS	Printed PDF re-measured: 4130 cut lines read back from this PDF match the model within 0.002 mm; 100 mm bars measure 100.00 mm

this bar must measure exactly 100 mm

cut — valley fold (towards you) — mountain fold (away) — glue tab: goes under the edge with the same number

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

1 STICK, THEN CUT

Glue each printout onto board with the printed side up. The printed side becomes the INSIDE of the helmet, so the outside stays clean for paint. Cut every solid line, including the three outer sides of each grey tab, the eye and mouth slots and the pivot holes.

2 SCORE AND PRE-FOLD

Dashed lines are valley folds: run an empty ballpoint along them on the printed side and fold towards you (printed side inside the fold). Dash-dot lines are mountain folds: score lightly with the knife (about a third of the way through) and fold away from you. Fold the base of every tab.

3 GLUE TABS

Every seam has a number printed on both pieces. The piece with the grey tab goes UNDER the edge with the same number: glue the tab, slide it under, press from outside until set. Edges meet flush.

4 SHELL STRIPS

Join S-C-1, S-C-2 and S-C-3 into the centre strip (front at S-C-1). Add S-R1 and S-L1 strips on each side, then R2/L2, then R3/L3, working front to back. Each strip's tabs lie along its inner edge. Finish with the side plates S-SIDE-R and S-SIDE-L.

5 TRY THE SHELL ON

Before going further, put the shell on. There should be about a finger's width all round. The front edge sits just above your brows.

6 FACEPLATE

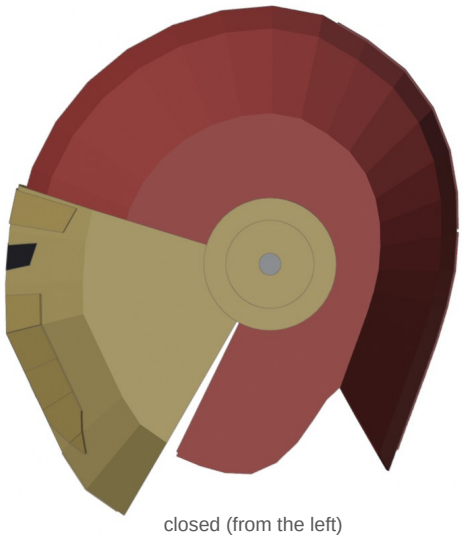
Build F-C, then F-R1 and F-L1, F-R2 and F-L2 (the eye slots line up across the seam), F-R3 and F-L3, then glue the pivot arms F-ARM-R and F-ARM-L onto the outer strips.

7 PIVOTS

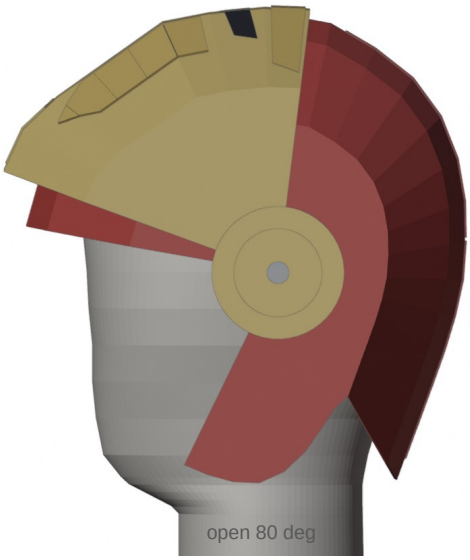
Glue an E-SIDE disc outside each side plate and an E-DISC outside each arm, holes aligned; add two E-RINGs on each E-DISC. Push a Chicago screw from inside through side plate, E-SIDE, washer, arm and E-DISC, and screw on the cap inside the rings. Tighten until the faceplate stays where you put it.

8 OVERLAYS AND FINISH

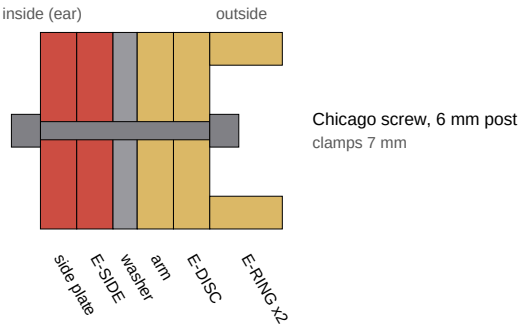
Glue O-BROW, O-CHEEK and O-CHIN onto the faceplate and O-CREST along the back centre strip (positions are shown on the next page). Seal with PVA or primer, then paint. Swing the faceplate fully open and closed to check nothing rubs.



closed (from the left)



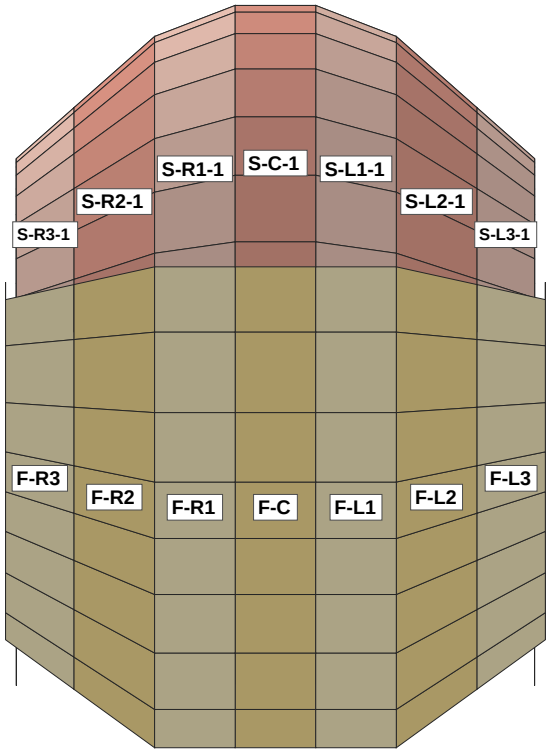
PIVOT STACK (section through the ear disc)



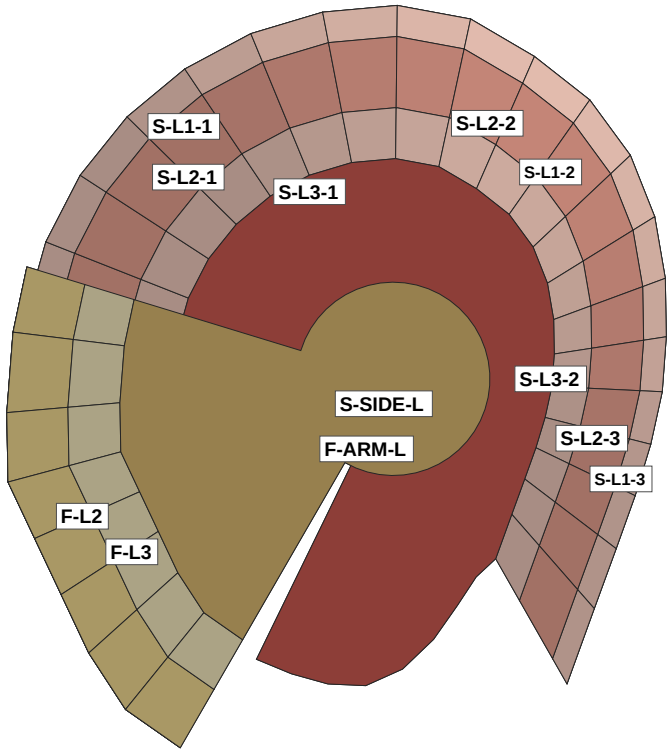
CARDBOARD HELMET - WHERE EACH PIECE GOES

Outside views of the finished helmet. Seam numbers on the stencils say exactly which edges meet.

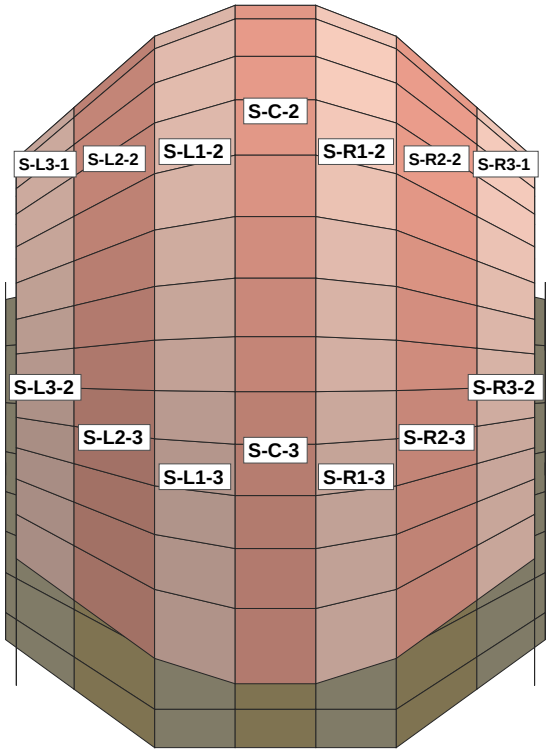
FRONT



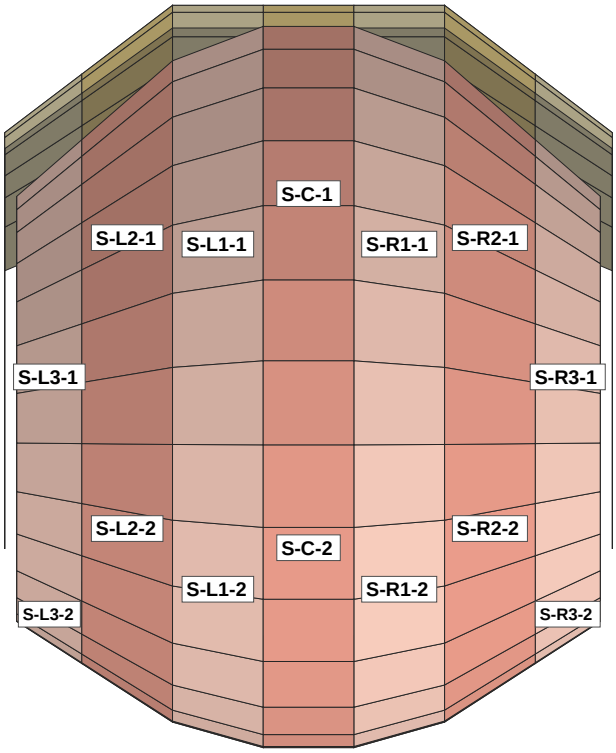
LEFT SIDE



BACK



TOP (front is down)



Overlays: O-BROW-L/R along the faceplate top band, O-CHEEK-L/R below the eye slots on F-L2/F-R2, O-CHIN at the bottom of F-C, O-CREST-1/2 on the back centre strip.

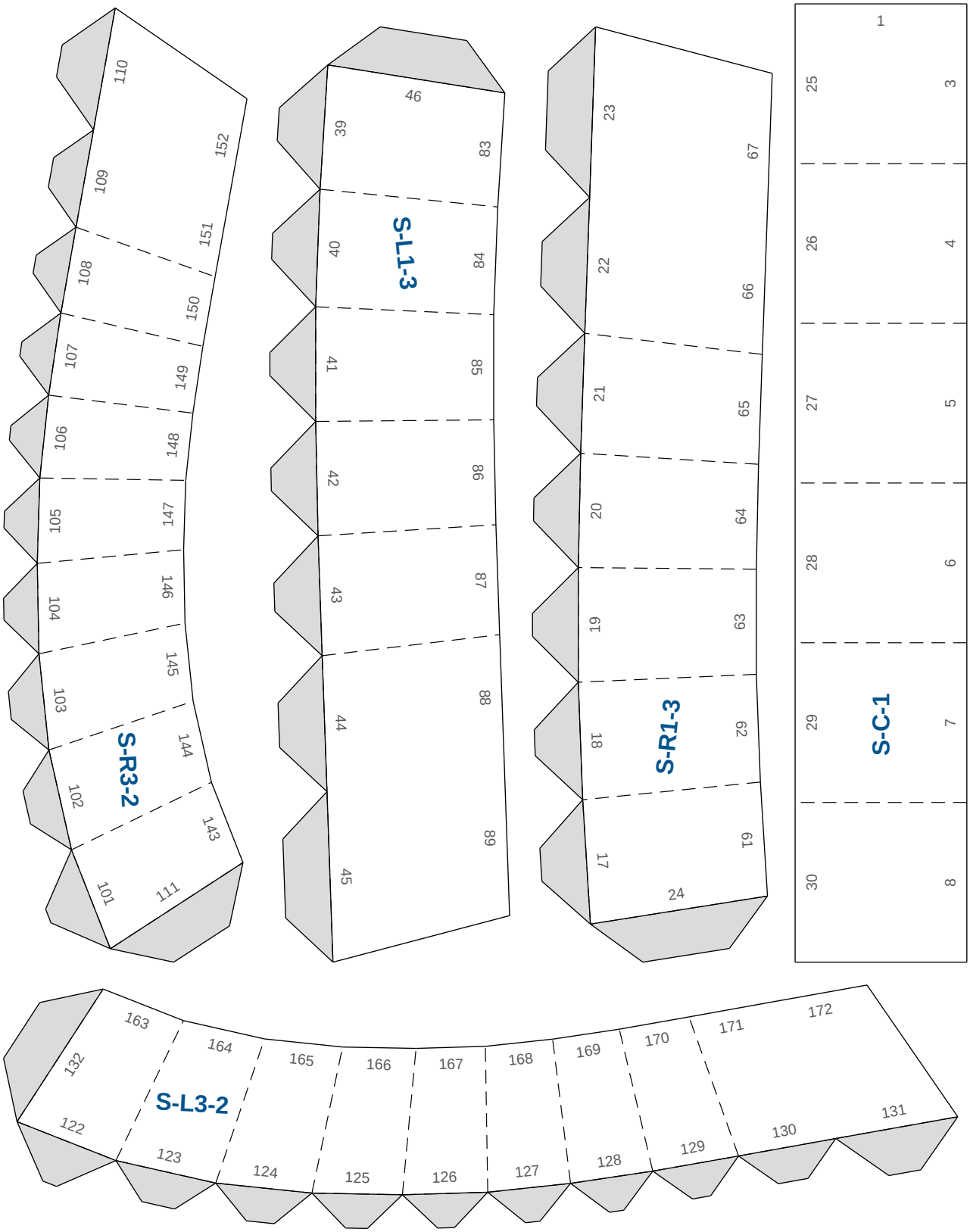
this bar must measure exactly 100 mm

cut — valley fold (towards you) — mountain fold (away) — glue tab: goes under the edge with the same number

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CARDBOARD HELMET - STENCILS

Pieces: S-C-1, S-R1-3, S-L1-3, S-R3-2, S-L3-2



this bar must measure exactly 100 mm

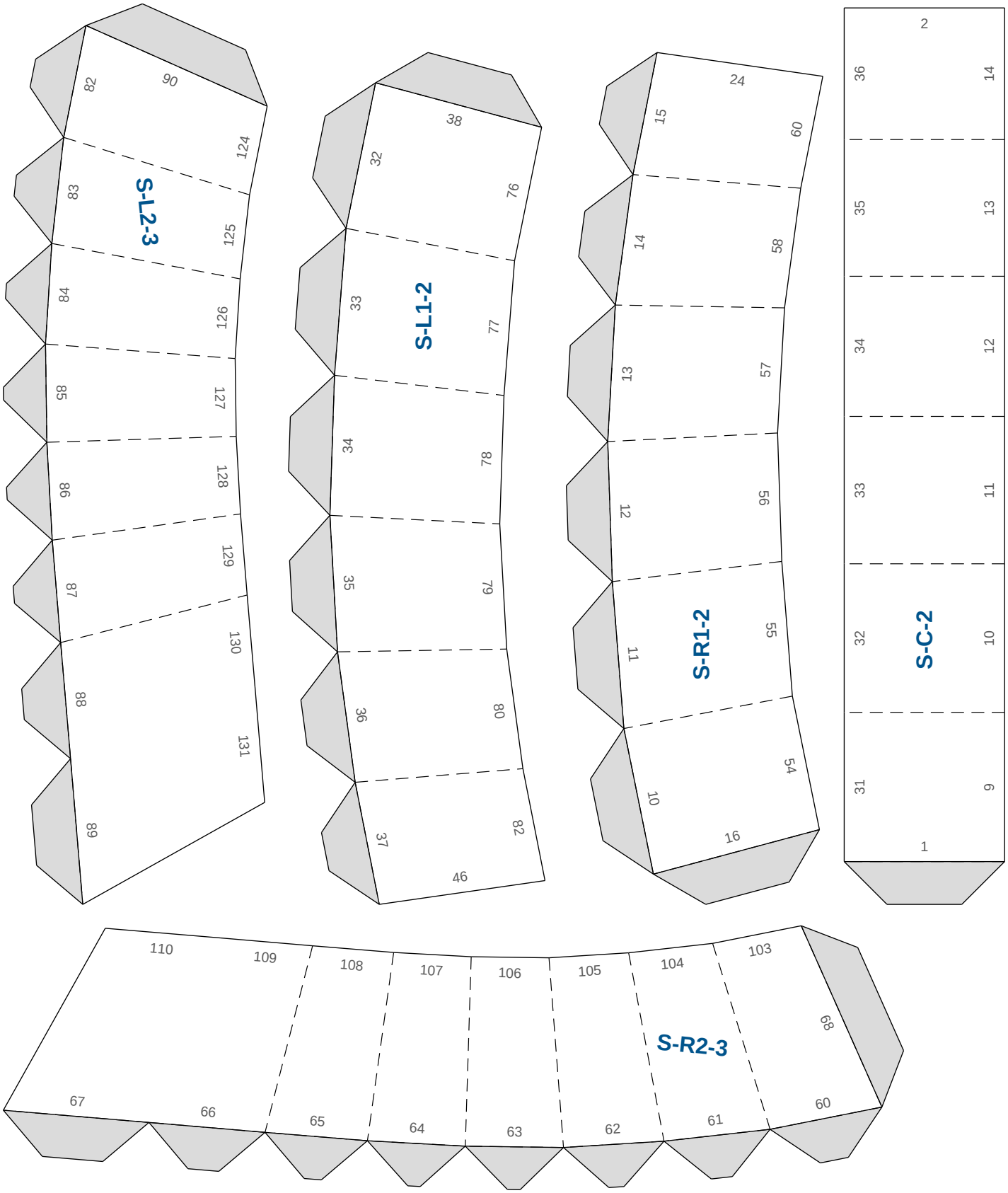
cut — valley fold (towards you) — mountain fold (away) — glue tab: goes under the edge with the same number

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-C-2, S-R1-2, S-L1-2, S-R2-3, S-L2-3



this bar must measure exactly 100 mm

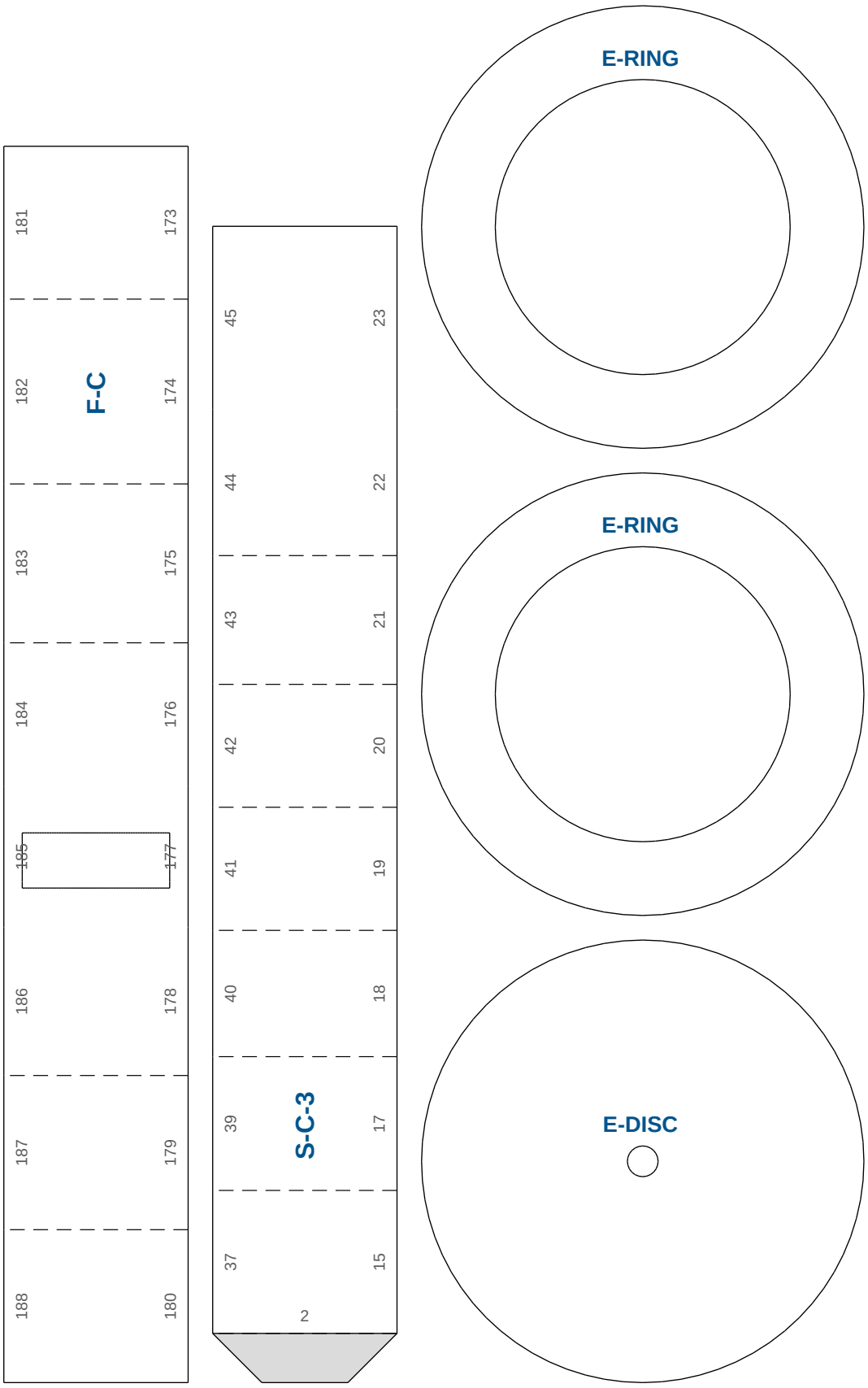
cut — valley fold (towards you) — mountain fold (away) — glue tab: goes under the edge with the same number

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-C-3, F-C, E-DISC (2/2), E-RING (1/4), E-RING (2/4)



this bar must measure exactly 100 mm

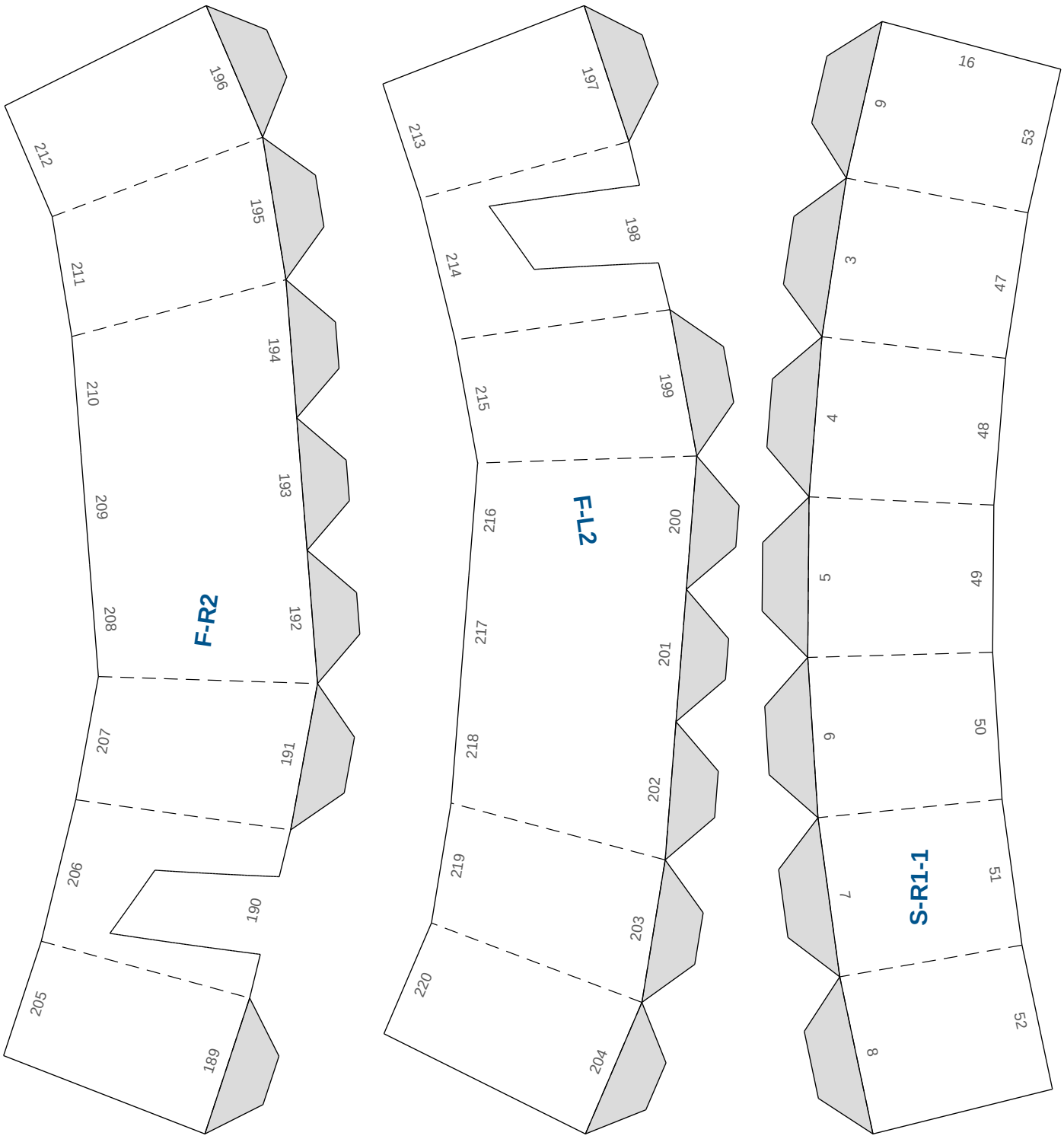
cut — — valley fold (towards you) — - — mountain fold (away) glue tab: goes under the edge with the same number

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-R1-1, F-R2, F-L2



this bar must measure exactly 100 mm

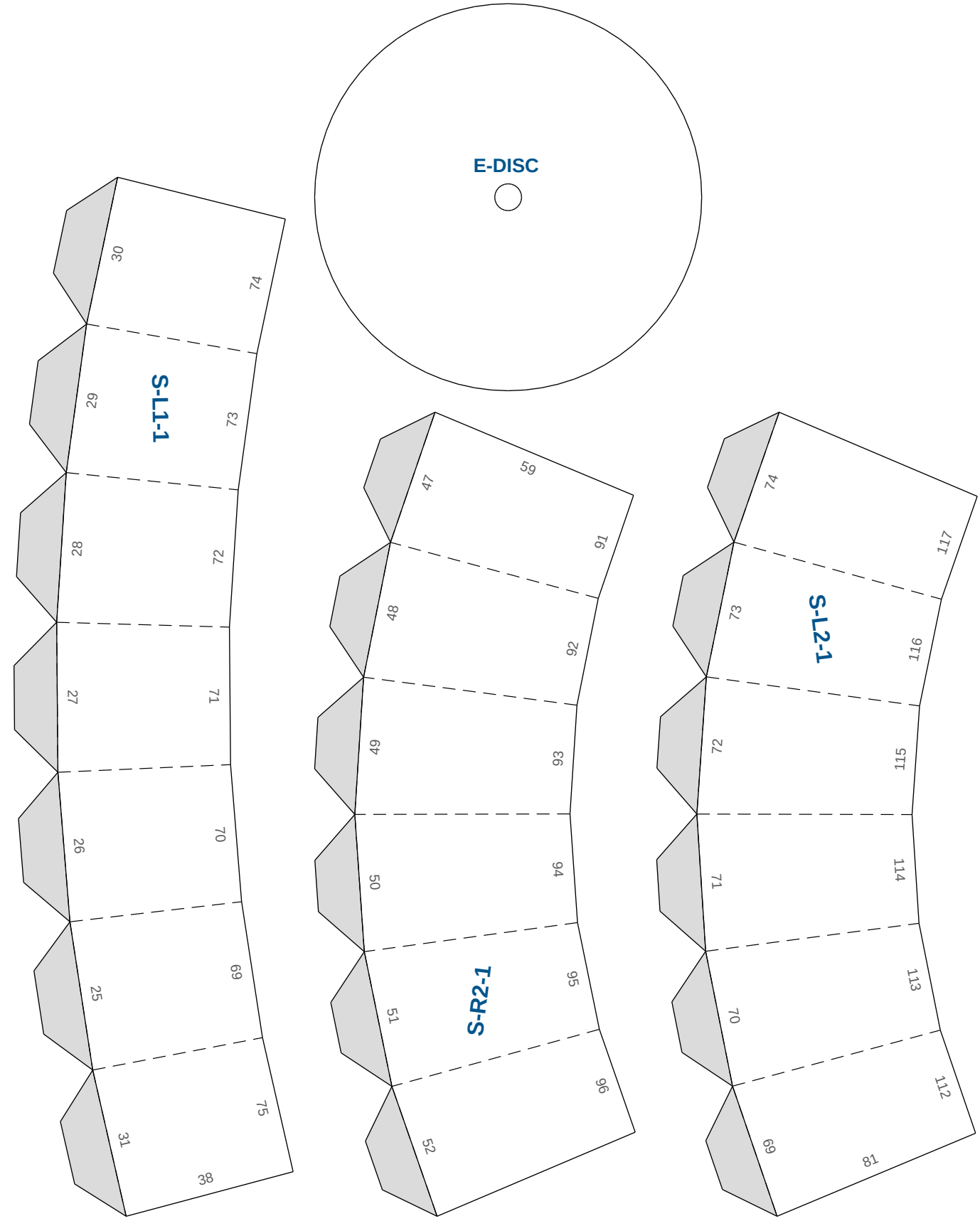
cut — valley fold (towards you) — mountain fold (away) — glue tab: goes under the edge with the same number

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-L1-1, S-R2-1, S-L2-1, E-DISC (1/2)



this bar must measure exactly 100 mm

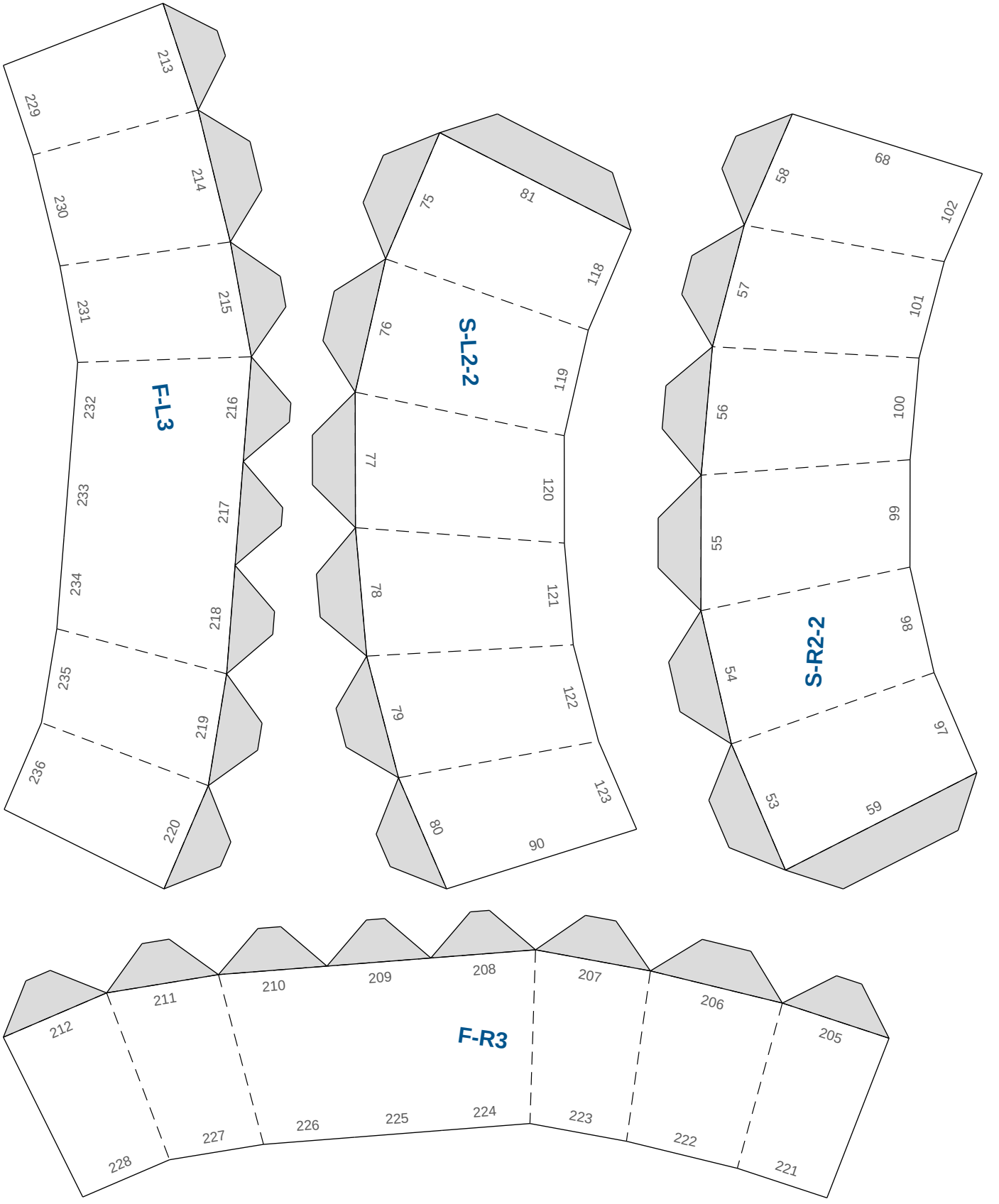
cut — valley fold (towards you) — mountain fold (away) — glue tab: goes under the edge with the same number

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-R2-2, S-L2-2, F-R3, F-L3



this bar must measure exactly 100 mm

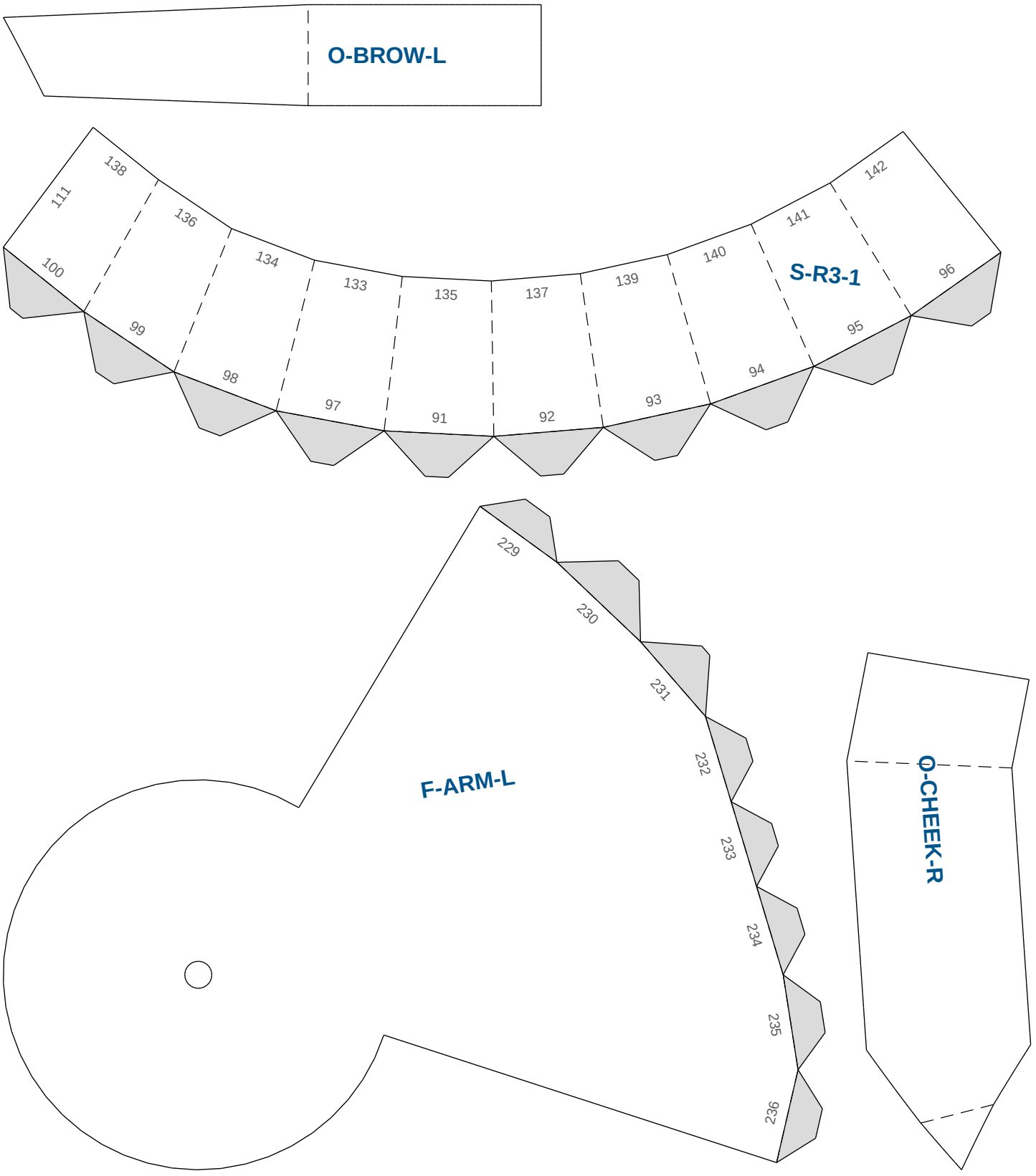
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Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-R3-1, F-ARM-L, O-BROW-L, O-CHEEK-R



this bar must measure exactly 100 mm

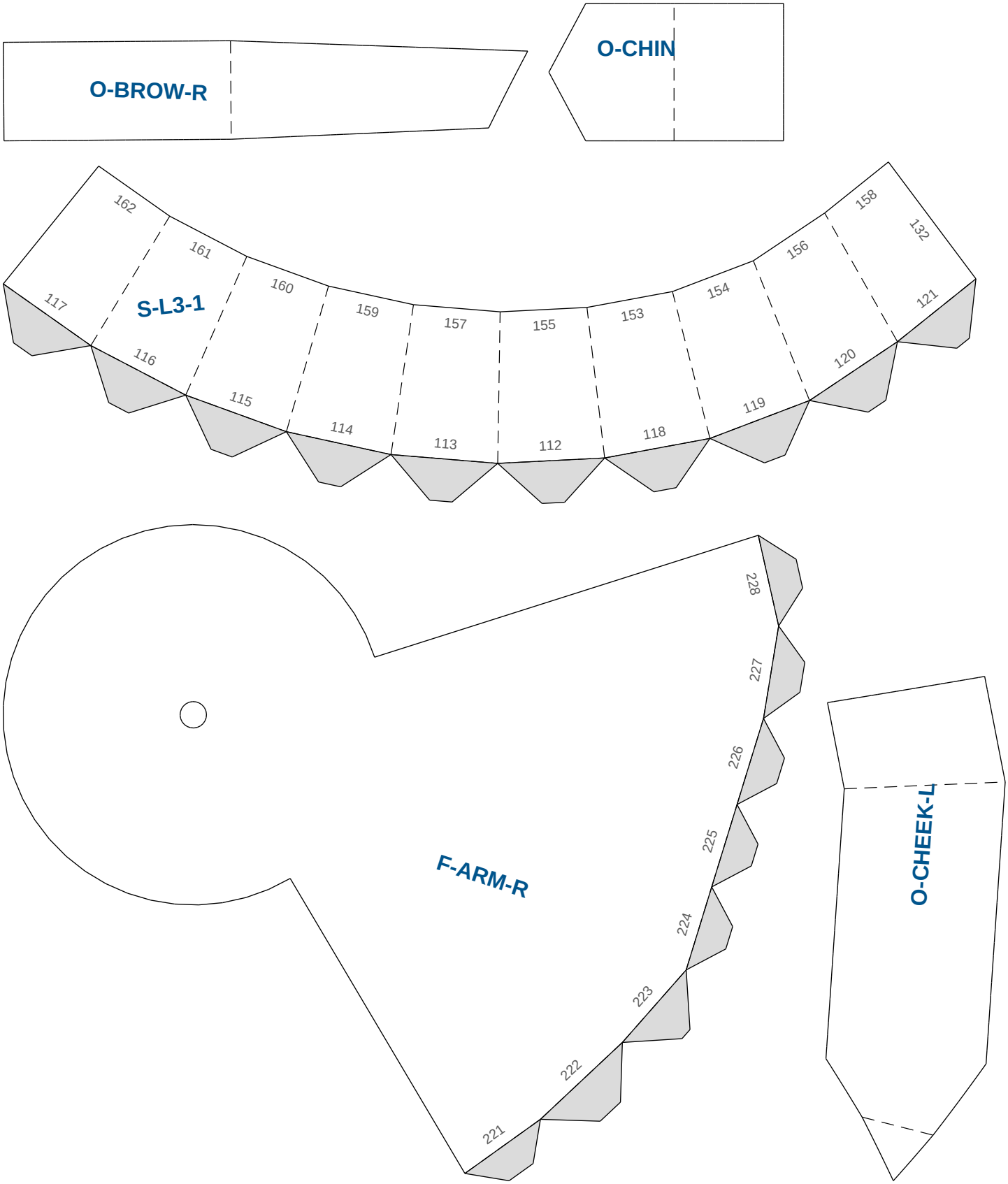
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Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-L3-1, F-ARM-R, O-BROW-R, O-CHEEK-L, O-CHIN



this bar must measure exactly 100 mm

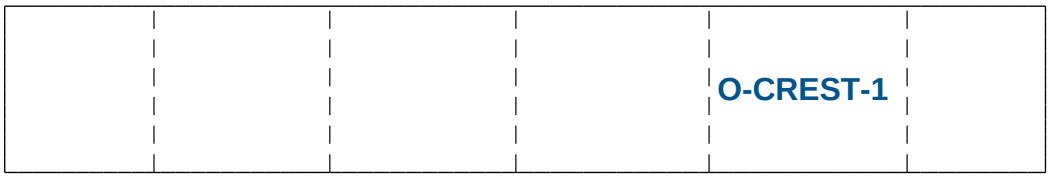
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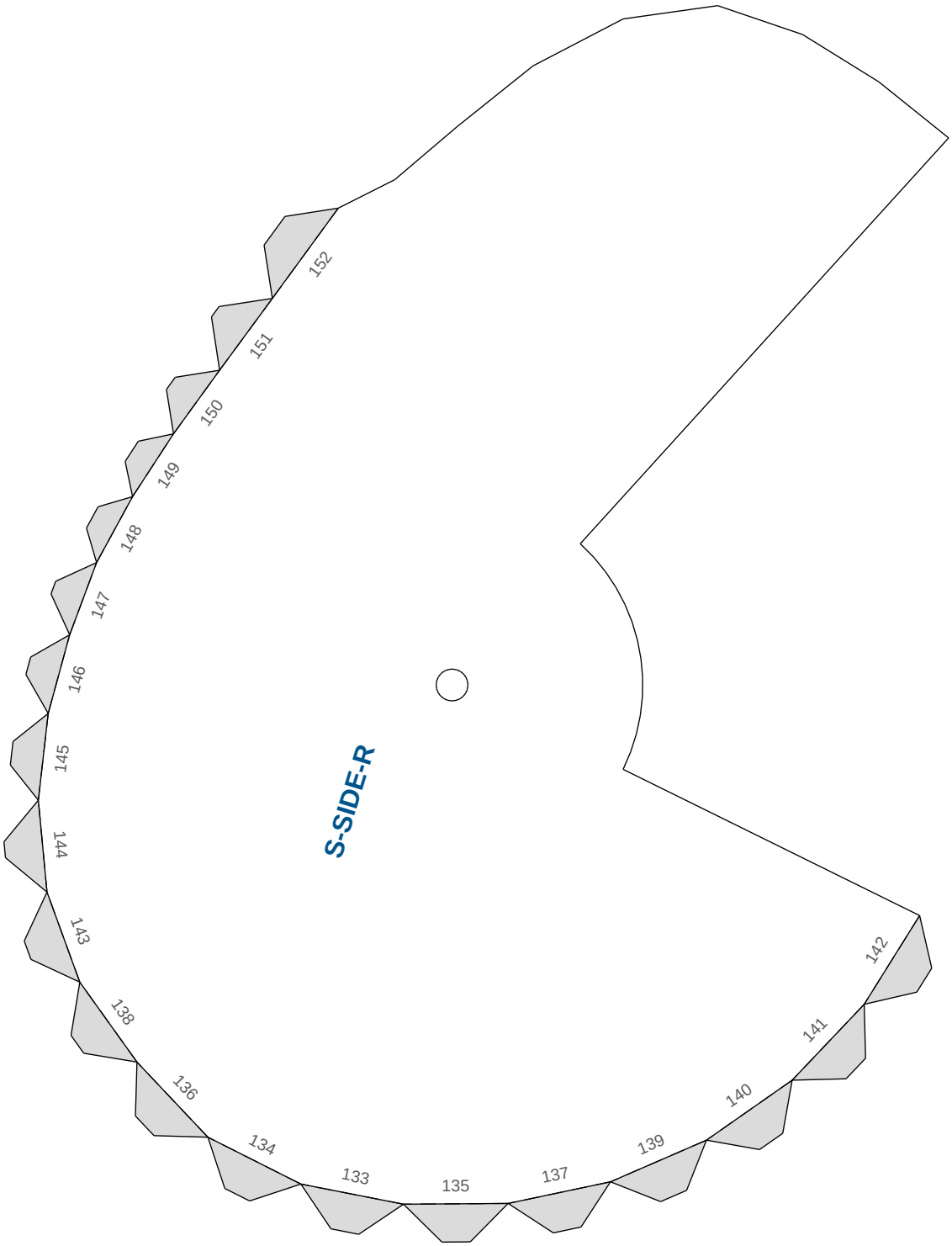
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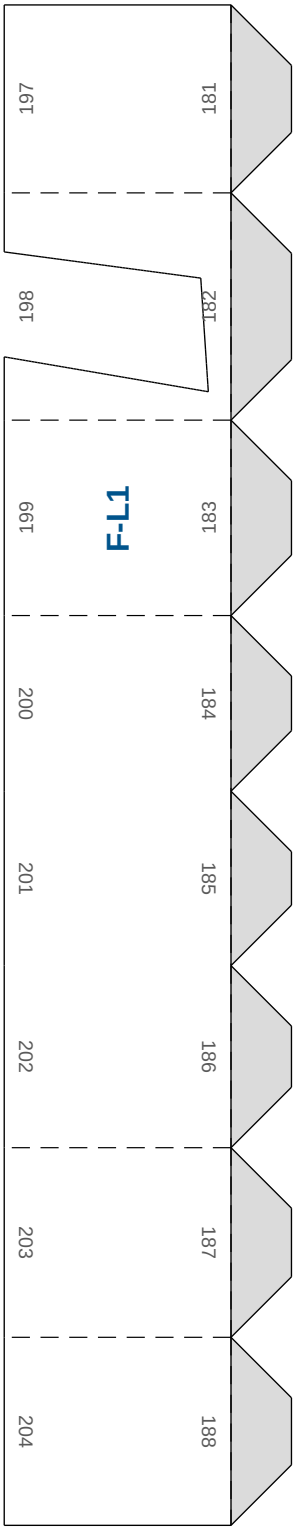
CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-SIDE-R, F-L1, O-CREST-1







this bar must measure exactly 100 mm

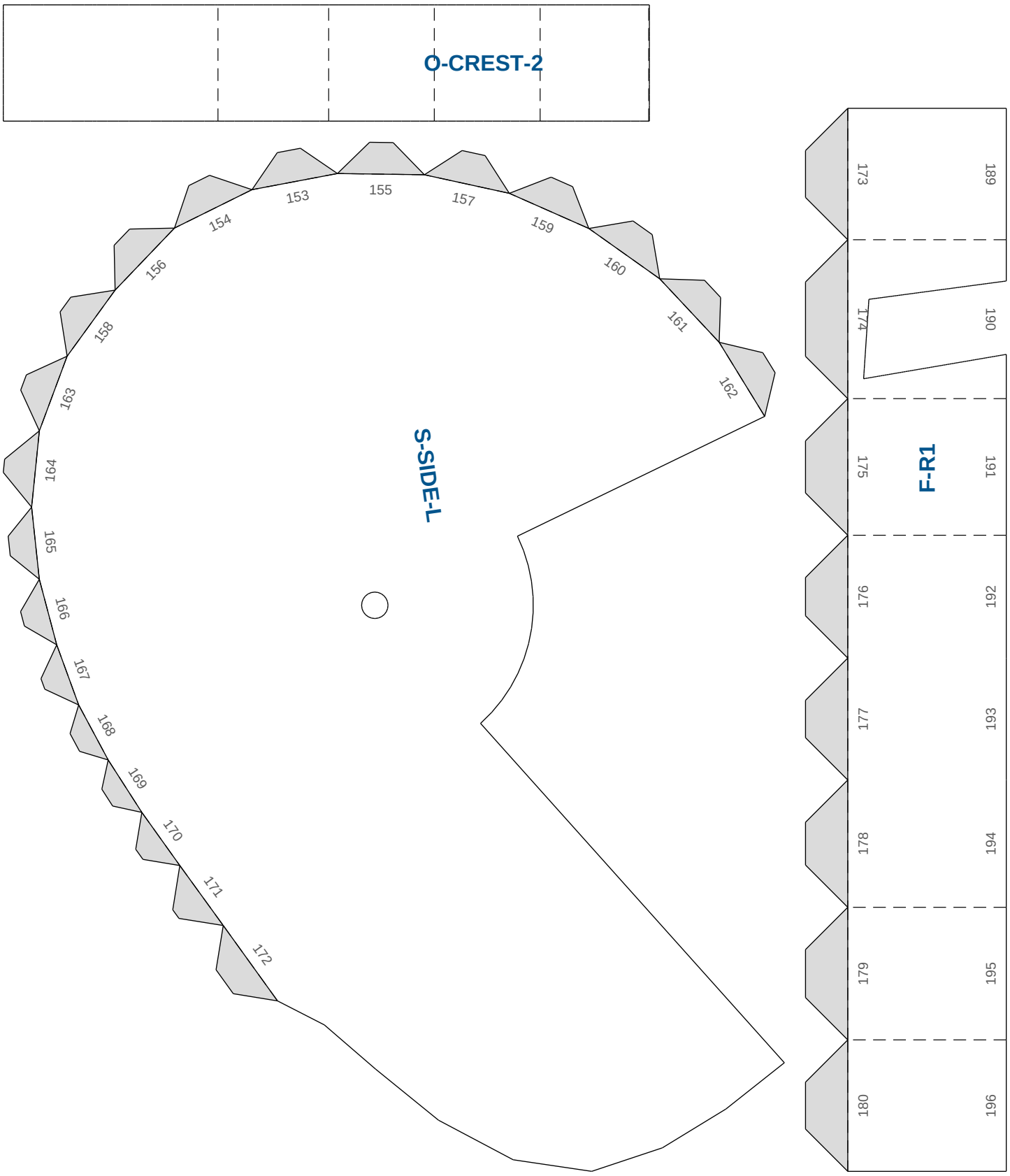
cut — valley fold (towards you) — mountain fold (away) glue tab: goes under the edge with the same number

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: S-SIDE-L, F-R1, O-CREST-2



this bar must measure exactly 100 mm

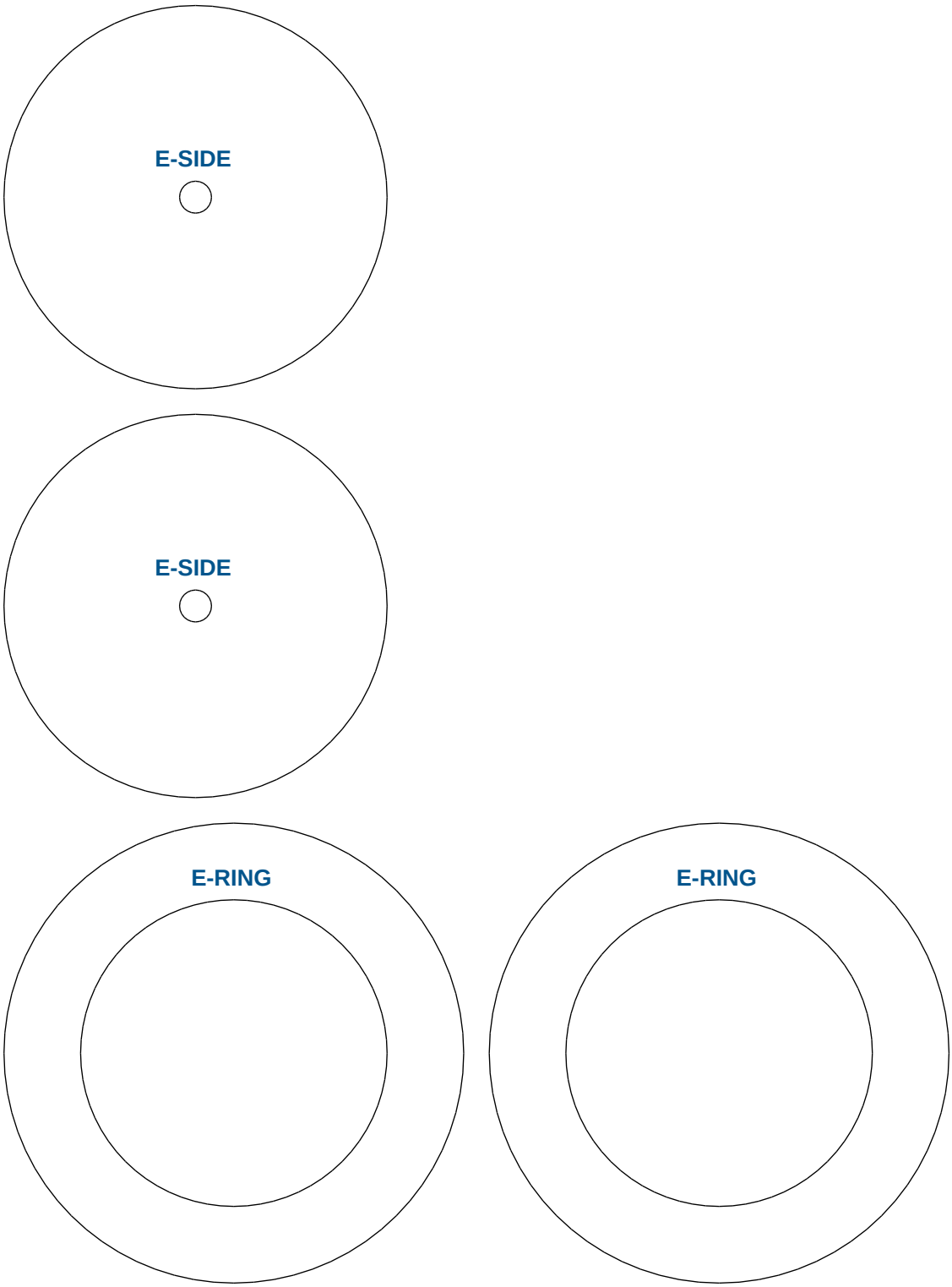
cut — — · valley fold (towards you) — · — mountain fold (away) glue tab: goes under the edge with the same number

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE

CARDBOARD HELMET - STENCILS

Pieces: E-SIDE (1/2), E-SIDE (2/2), E-RING (3/4), E-RING (4/4)



this bar must measure exactly 100 mm

cut — — · valley fold (towards you) — - — mountain fold (away) glue tab: goes under the edge with the same number

Print at 100% / Actual size. Printed side = INSIDE of the helmet.

CUT PATTERN - ACTUAL SIZE